

Lab 11

```
#include <ESP8266WiFi.h>
#include <ESP8266mDNS.h>
#include <WiFiUdp.h>
#include <ArduinoOTA.h>

// WiFi credentials
const char* ssid = "@";
const char* password = "123456789";

void setup() {
  Serial.begin(115200);

  // Built-in LED pin
  pinMode(LED_BUILTIN, OUTPUT);

  // Connect to WiFi
  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);

  Serial.println("Connecting to WiFi...");

  // Wait until connected
  while (WiFi.waitForConnectResult() != WL_CONNECTED) {
    Serial.println("Connection Failed! Restarting...");
    delay(5000);
    ESP.restart();
  }

  // Display IP address
  Serial.println("Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());

  // OTA setup
  ArduinoOTA.setHostname("esp8266");
  ArduinoOTA.setPort(8266);

  ArduinoOTA.begin();

  Serial.println("OTA Ready!");
}

void loop() {
  // Handle OTA updates
  ArduinoOTA.handle();

  // Blink built-in LED

  digitalWrite(LED_BUILTIN, LOW); // LED ON
  delay(500);

  digitalWrite(LED_BUILTIN, HIGH); // LED OFF
  delay(500);
}
```

